Adham (Leader – hardest/most central: OO design backbone + Java architecture + integration)

Part 0 — Foundation (everyone depends on this)

A0 – Classify Use Cases + Select the 12 Core Use Cases (4 simple / 4 moderate / 4 complex)

- **Use the use case diagram and classify all use cases into: simple / moderate / complex.**
- **Pick the final 12 (4 from each category).**
- **Output: classification table + list of 12 selected use cases (1–2 lines each) + rationale.**
- **Everyone depends on A0.**

A1 – Create the Deliverable #5 "Master Template" for the Technical Report (PDF structure)

- **Set report sections + numbering matching requirements.**
- **Output: report skeleton + where each diagram/table goes.**
- **Depends on A0.**

Part 1 — First-Cut Design Class Diagram (step-by-step transformation)

A2 – First-Cut Design Class Diagram (Global) + Partition into Layers

- **Transform from domain model to initial design classes (boundary/control/entity + data access).**
- **Define 3-layer architecture mapping (View/Controller/Domain/Data).**
- **Output: first-cut design class diagram (layered) + explanation steps.**
- **Depends on A0.**

A3 – Integration Gate: "Design Rules" for the team

- **Naming conventions, layer rules (no DB access from View/Domain), controller mediation rule, method signature style.**
- **Output: 1-page team design rules.**
- **Depends on A2.**

Part 2 — Use Case Realizations (hardest ones + final merge owner)

A4 – Complex Use Cases: Multi-layer Sequence Diagrams (for 4 complex UCs)

- **For each complex UC: sequence diagram including view objects + controllers + domain + DAO.**
- **Output: 4 sequence diagrams + identified methods/messages.**
- **Depends on A2/A3.**

A5 – Update Design Class Diagram (Final Merge Owner)

- **Collect methods from:**
  - **CRC (simple)**
  - **Communication diagrams (moderate)**
  - **Sequence diagrams (complex)**
- **Merge into final complete Design Class Diagram representing all 12 UCs.**
- **Output: final design class diagram (methods included) + change log summary.**
- **Depends on A4 + Andrew tasks (B*) + Ahmed tasks (C*) + Yassin tasks (D*).**

A6 – Package Diagram for Final Design

- **Group classes into packages (e.g., ui, controller, domain, dao, util).**
- **Output: package diagram + rationale.**
- **Depends on A5.**

Part 3 — Implementation (Walking Skeleton + 3-layer + DB)

A7 – Model-to-Code Transformation (UML Tool) + Walking Skeleton Setup

- **Generate Java code from final design class diagram (or partial skeleton early, then update).**
- **Create compile-able skeleton project structure.**
- **Output: generated code + cleaned project structure.**
- **Depends on A5 (or start early using A2 skeleton then update).**

A8 – Implement the 2 Simple Use Cases (DB interaction) – You implement the hardest/most central plumbing

- **You own:**
    - **DB connection strategy (SQLite/MySQL—whatever your course expects)**
    - **DAO base patterns**
    - **Controller mediation**
    - **Integrating with D4 UI screens**
- **Output: working end-to-end flow for 2 simple UCs + CRUD in DAO + UI triggers.**
- **Depends on D4 UI availability + B/C tasks for CRC/diagrams if they affect methods.**

A9 – Technical Report Integration (Final PDF Owner)

- **Merge all tables/diagrams and write rationales.**
- **Ensure "step-by-step transformation" is shown.**
- **Output: final PDF draft + final PDF.**
- **Depends on all.**

Andrew (diagram-heavy moderate/simple contributions + CRC + communication diagrams)

Part 1 — Simple Use Case Design (CRC)

B1 – CRC for 2 Simple Use Cases (with view + data layer classes)

Do CRC cards (PowerPoint is fine) for 2 simple UCs.

Must include boundary/view + controller + domain + DAO responsibilities.

Output: CRC slides + list of discovered methods.

Depends on A0 + A2/A3.

B2 – Update First-Cut Diagram Partition Notes (Feedback to Adham)

Validate that chosen classes match CRC findings.

Output: change suggestions list.

Depends on B1.

Part 2 — Moderate Use Case Design (Communication Diagrams)

B3 – Communication Diagrams for 2 Moderate Use Cases

Multi-layer communication diagrams: view objects → controller → domain → DAO.

Output: 2 communication diagrams + method/message list.

Depends on A0 + A2/A3.

Ahmed (system/data focus: DB-heavy UCs, DAO correctness, communication diagrams)

Part 1 — Moderate Use Cases (DB/system-heavy)

C1 – Communication Diagrams for 2 Moderate Use Cases (system/DB heavy)

- Choose moderate UCs with database CRUD and validations.
- **Output:** 2 communication diagrams + method list.
- Depends on A0 + A2/A3.

Part 2 — Implementation Support (Data Layer owner)

C2 – DAO + Database Schema Draft (aligned to design class diagram)

- Propose tables + primary keys + relationships + CRUD mapping.
- **Output:** ERD/table schema + DAO method signatures.
- Depends on A2; should be consistent with A5 eventually.

C3 – Support Adham in implementing 2 simple DB use cases

- Implement DAOs + SQL queries + test data seeds.
- **Output:** DAO classes + tested CRUD.

- Depends on A8 direction + C2.

Yassin (quality control + coordination: Jira/GitHub + validation + diagram consistency)

Part 0 — Project Management

D1 – Jira Timeline Setup + Dependencies

- Create Jira epics aligned to:
    - Use case selection/classification
    - First-cut DCD
    - Simple CRC
    - Moderate communication
    - Complex sequence
    - Final DCD + package
    - Model-to-code
    - Implement 2 simple UCs
    - Final PDF + submission bundle
- **Output:** Jira board with tasks A*/B*/C*/D* and dependency links.
- Depends on this plan (can start immediately).

D2 – GitHub Organization + Submission Bundle Owner

- Repo structure:
    - /docs (PDF + diagrams exports)
    - /uml (tool project files zipped)
    - /java (code)
- **Output:** clean repo structure + submission zip(s).
- Depends on ongoing work; finalized at end.

Part 1 — Validation & Consistency

D3 – Consistency Check: 12 Use Cases vs Diagrams vs Design Classes

- Verify each of the 12 UCs has:
    - a realization artifact (CRC/comm/sequence)
    - methods reflected in final DCD
    - correct layering rules
- **Output:** checklist + issues list.
- Depends on A5 draft.

D4 – Final Diagram Review Pass (before PDF freeze)

- Naming consistency, method completeness, layer boundaries.
- **Output:** “ready to submit” sign-off or fix list.
- Depends on A9 draft.

Submission

Deliverables Checklist (what you must submit)

- **PDF Technical Report** (A9 owner, inputs from everyone)
- **UML project files** (zipped)
- **Final Design Class Diagram** (A5)
- **Package Diagram** (A6)
- **CRC artifacts** for simple UCs (B1)
- **Communication diagrams** for moderate UCs (B3 + C1)
- **Sequence diagrams** for complex UCs (A4)
- **Java Project**
    - model-to-code skeleton (A7)
    - **2 simple DB use cases implemented in 3 layers + D4 UI included** (A8 + C3)
- **Jira timeline** (D1) + **GitHub organized repo** (D2)